expansion of the SDE method. The series expansion is therefore not particularly informative on its own, but only in relation to its action on some particular distribution.

A simple first order example of a Langevin dynamics integrator is the method obtained by composing one of the Symplectic Euler variants with an Ornstein-Uhlenbeck step. To get a feel for how the expansion goes, let us work out its terms for such a splitting scheme in the case of a one degree of freedom model with unit mass: $H = p^2/2 + U(q)$. To be explicit, let us say our numerical method first solves exactly the OU process, then evaluates the impulse due to the force $-U'(q)$ and lastly drifts in position space according to the momentum. This method is denoted in our notation as $[\![\mathrm{OBA}]\!]$, where the evolution of measure in one step can be computed by writing

$$\exp(h\hat{\mathcal{L}}^\dagger_{[\![\mathrm{OBA}]\!]}) = \exp(h\mathcal{L}^\dagger_{\mathrm{A}})\exp(h\mathcal{L}^\dagger_{\mathrm{B}})\exp(h\mathcal{L}^\dagger_{\mathrm{O}}).$$

In this one dimensional example, we substitute

$$Z = \mathcal{L}^\dagger_{\mathrm{A}} = -p\frac{\partial}{\partial q}, \qquad Y = \mathcal{L}^\dagger_{\mathrm{B}} = U'(q)\frac{\partial}{\partial p},$$
$$X = \mathcal{L}^\dagger_{\mathrm{O}} = \gamma\left(\mathrm{Id} + p\frac{\partial}{\partial p}\right) + \frac{\sigma^2}{2}\frac{\partial^2}{\partial p^2},$$

into (7.18), where $\sigma = \sqrt{2\gamma k_{\mathrm{B}}\mathrm{T}}$. This gives us an expansion for the Kolmogorov operator for our discretized map

$$\hat{\mathcal{L}}^\dagger_{[\![\mathrm{OBA}]\!]} = S = \mathcal{L}^\dagger_{\mathrm{A}} + \mathcal{L}^\dagger_{\mathrm{B}} + \mathcal{L}^\dagger_{\mathrm{O}} + \frac{h}{2}\left([\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{B}}] + [\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{O}}] + [\mathcal{L}^\dagger_{\mathrm{B}}, \mathcal{L}^\dagger_{\mathrm{O}}]\right) + \ldots$$

It is clear that even writing out the $\mathcal{O}(h^2)$ terms considerably increases the complexity of the right hand side.

The commutators are best considered in relation to their action on a function (or rather, their action on a density) $\rho(q,p)$. For example, we may compute directly

$$\begin{aligned}[\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{B}}]\rho &= -p\frac{\partial}{\partial q}\left(U'(q)\frac{\partial \rho}{\partial p}\right) - U'(q)\frac{\partial}{\partial p}\left(-p\frac{\partial \rho}{\partial q}\right)\\ &= -p(U'(q)\rho_p)_q + U'(q)(p\rho_q)_p,\\ &= -pU''(q)\rho_p - pU'(q)\rho_{pq} + U'(q)\rho_q + U'(q)p\rho_{qp}.\end{aligned}$$

As this is the commutator of two 1st order differential operators, the mixed partial term vanishes, leaving

$$[\mathcal{L}^\dagger_{\mathrm{A}}, \mathcal{L}^\dagger_{\mathrm{B}}]\rho = -pU''(q)\rho_p + U'(q)\rho_q.$$

Similar computations give

$$[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho = \gamma p \rho_q + \sigma^2 \rho_{qp},$$
$$[\mathcal{L}_{\mathrm{B}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho = \gamma U'(q) \rho_p.$$

To first order, the discretization map corresponding to scheme ⟦OBA⟧ alters the phase space density at each step as if it were solving the time dependent PDE

$$\rho_t = \hat{\mathcal{L}}^{\dagger} \rho,$$

where

$$\hat{\mathcal{L}}^{\dagger}\rho = \mathcal{L}_{\mathrm{LD}}^{\dagger}\rho + \frac{h}{2}\left[-pU''(q)\rho_p + U'(q)\rho_q + \gamma p \rho_q + \sigma^2 \rho_{qp} + \gamma U'(q)\rho_p\right] + \mathcal{O}(h^2).$$

The validity of this sort of expansion, even for small h, should be considered carefully. Defining

$$\mathcal{L}_{\mathrm{LD}}^{\dagger} = \mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger} + \mathcal{L}_{\mathrm{O}}^{\dagger}$$

as the Kolmogorov operator for Langevin dynamics, the assumption of validity becomes more credible as we find that the operator $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ has a compact resolvent and therefore a discrete spectrum, i.e. simple eigenvalues, as discussed in the previous chapter. It will have one zero eigenvalue corresponding to the invariant density ρ_β and all the remaining eigenvalues lie in the negative half plane. Of course it is not just this operator whose properties are important, but also all the other operators arising in the expansion, and bounding these in relation to $\mathcal{L}_{\mathrm{LD}}^{\dagger}$ is not a simple task (particularly for higher orders).

7.7.3 Perturbation Technique

Suppose we are able to derive a series expansion for $\hat{\mathcal{L}}^{\dagger}$ of the form

$$\hat{\mathcal{L}}^{\dagger} = \mathcal{L}_{\mathrm{LD}}^{\dagger} + h\mathcal{L}_1^{\dagger} + h^2\mathcal{L}_2^{\dagger} + \mathcal{O}(h^3),$$

for some perturbation operators $\mathcal{L}_i^{\dagger}$. We assume that the invariant density has a similar expansion in powers of the stepsize, of the form given in (7.17). Inserting this ansatz into the stationary Fokker-Planck equation

$$\hat{\mathcal{L}}^{\dagger}\hat{\rho} = 0,$$